

# Designing for Black Health Futures: Exploring Black-Centered Design from a Human Factors Lens in Pursuits of Equitable Black Health Futures

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## Abstract

My research investigates the use of Afrofuturism, guided by human factors principles, to design digital health tools that address Black health equity. Rooted in Black-centered design (BCD), my work centers on incorporating the voices, values, and lived experiences of Black individuals in the design process. By integrating patient ergonomics, I aim to create a systematic approach to BCD, focusing on improving health equity for Black maternal mental health. My initial literature review examines design approaches used in Black digital health technologies, while my first field study identifies barriers, pain points, and resilience in accessing mental health resources for Black birthing individuals. Future research will include insights from designers and healthcare providers, leading to a co-design session leveraging Afrofuturism to develop digital tools that meet the mental health needs of Black birthing people. This work aims to establish a structured, human factors-based approach to BCD for health equity.

## CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • HCI design and evaluation methods;

## Keywords

Black-centered design, human-centered design, Afrofuturism, digital health equity

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## 1 Introduction

Health is recognized as a fundamental human right [15], yet racism remains a significant driver of persistent health disparities [16–19]. According to the World Health Organization, health equity means

ensuring that all individuals, regardless of their background, can achieve their full potential for health and well-being [15]. Healthy People 2030 defined health disparities as “health differences that are closely linked with social, economic, and/or environmental disadvantage which adversely affect groups of people who have systematically experienced greater obstacles to health based on their racial or ethnic group, religion, socioeconomic status” and other social determinants [20]. Despite limited race-based data, Canada ranks last among high-income nations in healthcare equity, as reported by the Commonwealth Fund [21]. Achieving health equity demands addressing systemic inequities with fairness and justice, particularly for vulnerable populations [22].

For Black communities, systemic racism and historical medical injustices have resulted in severe health disparities. These communities, including individuals of African and Afro-Caribbean descent, have faced generations of discrimination, underscoring the need for targeted, race-based research. Black-Centered Design (BCD) offers a promising framework for addressing these disparities by prioritizing the healthcare needs and lived experiences of Black individuals. This research explores BCD’s application to maternal mental health in Canada—a critical starting point due to its direct generational impacts on mothers and infants.

Black birthing individuals are at heightened risk for maternal mental health conditions, including postpartum depression and anxiety [23]. Estriple et al. found that Black mothers and birthing individuals are at a disadvantage in autonomous decision-making due to factors such as limited access to resources, lack of universal screening, disjointed healthcare systems, and inadequate mental health education [24]. While precise statistics are unavailable due to Canada’s limited race-based data, evidence highlights significant treatment disparities influenced by cultural beliefs, racial discrimination, and immigration status [25]. In the United States (US) approximately 1 in 5 new US mothers experience mental health concerns, with a staggering 40% of Black women reporting symptoms [26]. Alarming, nearly half of these women do not receive any treatment or support for these conditions, underscoring the urgent issue faced by Black birthing communities. Digital mental health technologies hold promises in reducing access barriers and stigma within birthing communities [27]. However, poorly designed technologies risk exacerbating inequities. This research investigates the potential of BCD, combined with patient ergonomics, to address the Black maternal health crisis and foster equitable health outcomes.

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## 2 Background and Related Work

Due to the persistent nature of health disparities among Black communities, addressing these disparities require an innovative and inclusive approach that prioritizes their lived experiences and needs [1]. As technology becomes increasingly integral to health-care delivery, it is essential to not only consider but “center the needs, values, and desires of Black bodies and lives” in the design and development of new technologies [2]. Black-centered design (BCD) emerges as a transformative framework that places the voices, values, and lived experiences of Black individuals at the forefront of the design process. Industrial engineer Woodrow W. Winchester III began to explore how BCD and Afrofuturism offers a pathway towards more equitable healthcare for Black communities. Winchester defines BCD as an approach that uses the nuanced complexities of Black identity as a guiding ethos for creating more equitable and just technological tools and systems [2]. The core principles of BCD include [3]: 1) engaging design principles that explicitly foreground Black lives and bodies, 2) seeking opportunities to involve the Black community in technological design, and 3) embracing interdisciplinarity in design practices [3]. BCD represents a way of thinking that integrates Black culture, historical context, and lived experiences within the current paradigm to dismantle systemic inequalities.

### 2.1 Design and Solutionism

In comparison to Human-centered design (HCD), which focuses on making systems usable and useful by centering users’ needs and requirements [4], BCD offers a more inclusive, culturally specific approach that reimagines mainstream technological practices for marginalized populations. Scholars have explored BCD across various domains regarding financial and economic concerns, Science Technology Engineering and Mathematics (STEM) education, as well as just the overall futures of Black individuals [5–7]. In many of these cases, BCD is applied using the design lens of Afrofuturism, which envisions empowered futures for Black lives by centering justice and reimagining possibilities within the design process [8]. Scholars such as Christina Harrington and Winchester III have begun to investigate the potential of BCD and Afrofuturism towards health equity outcomes [7, 9]. This research has shed a light through the envisioning of health equity from a new standpoint, where Black communities are encouraged to look outside of the box of the traditional offerings to address their health needs.

While BCD does not present itself as a definitive design solution, it aims to catalyze more inclusive design thinking and decision-making when addressing the needs of the Black community. This is especially important given critiques of “techno-solutionism” within the design process, which can lead to superficial fixes that fail to address underlying issues [10, 11]. Ruha Benjamin highlights this concern stating that although the ethos of design matters, we must continue to ask ourselves, “In the breathless race for newer, faster, better technology, what ways of thinking, being, and organizing social life are potentially snuffed out? If design is treated as inherently moving forward, that is, as the solution, have we even agreed upon the problem?” [12]. This leads us towards the question of how can we better understand the problem at hand without putting all of our efforts in a predetermined solution?

### 2.2 Design and Human Factors

One critical gap within understanding the problem lays from the health domain lies within accounting for patient work within the process of attempting to pursue health resources and support. Human factors and ergonomics have been a discipline focused on understanding how we can improve human work in various sociotechnical systems [13]. More recently, patient ergonomics has been a growing discipline within human factors research that analyzes the effortful work that patients, caregivers, and non-clinical support put in achieving their intended health goals [14]. In essence, it focuses on improving patients’ performance of effortful activities in pursuit of health goals [13]. While patient ergonomics has grown over the years, there is a gap in the use of this method as it pertains to investigating marginalized groups, patient experience, and wellness promoting behavior [13].

## 3 Research Methods and summary

Although scholars have begun to substantiate the need for BCD approaches, few have focused on how we can systematically implement and evaluate its effectiveness within the health domain. This research aims to explore the implementation and evaluation of BCD through the integration of patient ergonomics to improve health equity outcomes for Black people. By engaging with Black communities and leveraging interdisciplinary insights from designers, healthcare professionals and community members I plan to develop best practices for the implementation of interventions that resonate with the unique challenges and strengths of Black individuals. Ultimately, this research advocates for a shift toward more inclusive and equitable design practices that contribute to the broader goal of health equity. The following outline the research questions that guide my dissertation work:

- **RQ1:** What is the current state of design regarding designing technology and tools to meet the needs of black people who experience health inequity?
- **RQ2:** What are the current failures, gaps, and resiliencies within mental health care for black birthing communities?
- **RQ3:** How have people in the field (designers and community health providers) designed health tools and systems to meet the needs of black community members?
- **RQ4:** What unique contributions can Black-centered design offer to health equity? How can we use patient ergonomics to inform BCD through the lens of patient work? How can we evaluate the impact of this design framework?

### 3.1 Objective 1: Synthesis of existing literature on design approaches used to address Black health equity.

To address the racial disparities faced by Black communities, this research aims to understand the current literature of designing digital health technology (DHT) for health equity outcomes. I have conducted a comprehensive scoping review of existing literature. The focus is on identifying design best practices that contribute to Black health equity. Specifically, I analyzed the types of DHT used, the design methodologies applied, the specific health problems addressed, and the resulting health impact. This review offers

a snapshot of the current state of design practices in promoting Black health equity and a robust foundation for compiling design best practices. Articles were gathered from various databases such as PubMed, Scopus, and ACM. In total 1574 articles were retrieved from the search. Title/abstract and full-text screening was conducted with three screeners inclusive of myself. Twenty articles met inclusion criteria and were moved to the data extraction stage. Data extraction has been completed and results were thematically analyzed. Summary of results have been used to draft a manuscript to be submitted as a conference paper for CHI 2025.

### 3.2 Objective 2: Use of patient centered cognitive task analysis to identify critical incidents impacting mental healthcare access for Black birthing communities.

The highlighted disparities emphasize the need to identify the failures, gaps, and resiliency within the mental health care process that have been impacting Black birthing people. Recognizing that different health groups have distinct requirements; this objective focuses on Black birthing individuals as a case study. These individuals currently face disparities in both mental and physical health outcomes. This study is currently underway through a phenomenological study—immersing in the lived experiences of Black birthing people. I aim to identify pain points and challenges they encounter while navigating mental healthcare systems and tools. To achieve this, I am conducting both surveys and semi-structured interviews with 10-12 new mothers. To ensure comprehensive insights, I will purposefully oversample Black birthing individuals. By comparatively analyzing critical events among both Black and other birthing individuals, I seek to explain disparities observed in the literature. I will analyze the data using a patient centered cognitive analysis approach to map and identify critical incidents impacting access to mental health support based on race and develop personas based on these findings.

### 3.3 Objective 3: Conduct an ethnographic study focused on assessing the barriers and facilitators to BCD from the perspective of community health providers, patients, and designers.

To develop a framework that can be implemented for BCD in the health domain, we must embrace interdisciplinarity through the engagement of providers, patients, and designers to understand the barriers and facilitators to this work, especially in a Canadian context where race-based health data is emerging. To address this research question, I plan to host two focus groups discussions with designers who design tools and systems for the health needs of Black people. Second, I plan to interview approximately 5-8 Black community health provider to understand their vision for designing for Black health equity. Both focus group discussions (FGDs) and interviews will focus on understanding barriers, facilitators, and their perspectives on design best practices for Black health equity. By gathering information from both Black design practitioners and community health providers, I will gain a deeper understanding of how Black communities design for their own health needs. This

triangulation of knowledge—from literature, professional designers, and community perspectives—will form a robust foundation for an adapted BCD framework that offers design best practices/considerations for achieving health equity outcomes. To identify potential collaborators for this workshop, I attended the State of Black Design conference that was hosted at University of North Texas in March 2024. This conference helped to deepen my knowledge of the current stage of Black centered design from various disciplines such as healthcare, education, architecture, and more. Lastly, I am developing Black community health center collaborations within Canada through partnerships with Durham Community Health Center and Women's Health in Women's Hands. I am currently beginning the ethic review process for this study.

### 3.4 Objective 4: Pilot study implementing Black-centered design framework that is adapted to encompass patient work and evaluate outcomes.

While this work is not solutionism oriented, I plan to leverage insights from previous objectives to inform an adapted BCD framework that accounts for patient work and health equity impact. This framework will draw from best practices identified in the literature and input from key stakeholders, while integrating requirements from Black people facing inequity. Using the adapted BCD framework, I will conduct a co-design session with designers, providers, and Black birthing people focused on developing an intervention for addressing the mental health needs of Black birthing individuals. Impact will be accessed using the developed framework. The findings from the objective will help to evaluate the impact of the adapted BCD framework within the health domain and begin to substantiate Black-centered design within the health domain. The outcomes will be evaluated through the expected outputs gathered from previous studies.

## 4 Challenges and Expected Outcomes

While I have made significant progress on my research, several critical questions remain as I work towards my goals. One central question is how to evaluate BCD approaches, such as Afrofuturism, without diminishing their inherently speculative nature. A key question that lingers is: *Is it possible for speculative design to coexist with practical, measurable outcomes, especially within the realm of healthcare?* I also find myself reflecting on how to develop Objective 4 in a way that not only envisions future possibilities but also provides concrete steps toward realizing those futures. These questions are pivotal because I aim to do more than imagine a better future for Black communities—I want to create a pathway for these communities to achieve it.

By grounding Black-Centered Design (BCD) frameworks in real-world applications, I aim to provide both inspiration and practical benefits to the communities I seek to serve. The intended contributions of my graduate work are threefold: 1) Offer empirical findings on how BCD can be leveraged and operationalized within the health domain to achieve equitable outcomes. 2) Integrate patient ergonomics into the design process, ensuring a holistic and inclusive approach to design. 3) Evaluate the impacts of BCD on advancing health equity through systematic assessment. Best practices for

evaluation will be identified through a comprehensive literature review and insights gathered from designers. Additionally, integrating findings from interviews with Black birthing individuals will underscore the critical patient work involved in maintaining and achieving mental health care. Finally, through my co-design study, I plan to highlight the structured methodology used to develop and assess the outcomes of the BCD approach. These insights will provide valuable guidance for enhancing equity-driven design practices in healthcare.

## 5 Research Background

Throughout my academic career, I have focused on advancing the mental and physical well-being of Black communities through a human factors lens. I am eager to contribute to the CHI 2025 Doctoral Consortium by bringing my interdisciplinary perspective, which integrates human factors engineering, psychology, and healthcare, all framed through a social justice lens. While I have long explored the use of technology in my work, delving into Afrofuturism and user-centered design has sparked a profound interest in the human-computer interaction (HCI) space. Although I am relatively new to this field, I am enthusiastic about the opportunity to both learn from and contribute to this vibrant community. My academic journey has been profoundly shaped by HCI and human factors research, particularly insights drawn from CHI proceedings. Participating in the consortium would mark my first opportunity to engage in such a venue—one that aligns seamlessly with my research focus. Through the doctoral consortium, I aim to collaborate with peers and faculty to refine my research on Black-centered design for digital health equity. Specifically, I hope to generate actionable ideas that incorporate HCI principles and perspectives into my work, which is grounded in the foundational research of esteemed HCI scholars. I am particularly eager to use this opportunity to critically address the lingering questions and challenges inherent in my research, fostering innovation and clarity.

My long-term aspirations fluctuate between a career in academia and entering the industry to focus on health technology design. I am still evaluating which path aligns best with my goals. Teaching, education, and continuous learning have always been my core passions, which naturally draws me towards becoming a professor and eventually starting my own research lab. While this remains a strong interest, I am also intrigued by the field of implementation science and how I can bring some of my visions to life in the industry. Ultimately, I want to find a balance between these two avenues—combining the exploration and dissemination of knowledge in academia with the practical, impactful application of ideas in industry.

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